

INNOVATION

Key Innovation 1: Making the care-establishment pathway computable. Phase IIB modeled the front end of eldercare planning: household needs and means, likely benefits and aid, and appropriate services and providers. The CRP addresses the harder engineering problem that follows. Eldercare unfolds through a finite but highly variable set of real-world entities, documents, communications, decisions, and delays that differ by household, program, provider, and geography. Reliable automation first requires a computational representation of that system.

We therefore propose a longitudinal **Care Establishment Model** organized around seven eldercare-specific domains: needs, means, funding, service, execution, capacity, and outcome. Each domain contains more granular substates developed during Aim 1. Natural language, uploaded documents, calls, emails, forms, portal responses, and provider observations are interpreted by the Phase IIB eldercare LLM and normalized into this common state. The reverse process assembles the household's verified state into the program- or provider-specific information needed for a bounded workflow. Applications, requests for documentation, provider denials, waiting lists, service starts, and disruptions become timestamped events with geography and provenance rather than disappearing into disconnected inboxes and phone calls.

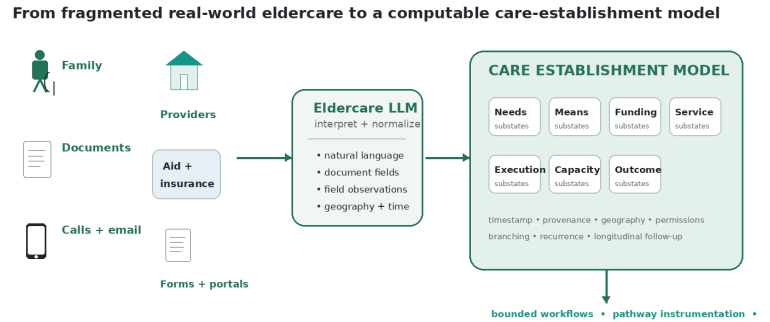


Figure 5. Fragmented eldercare inputs are normalized into a longitudinal Care Establishment Model that drives execution and measurement.

This is the load-bearing technical advance for the rest of the CRP. A software system cannot reliably automate a fragmented pathway unless it can observe what state the case is in, what changed, who owns the next action, and what evidence defines completion. The model also makes the pathway measurable. Static directories describe what should exist and claims describe what was billed. Executed cases reveal what actually happened between recognized need and aid or care becoming established. Over time, Olera can therefore observe where, why, and how quickly care establishment fails at household and geographic levels.

Key Innovation 2: AI agents that execute and learn from the care-establishment pathway. The prevailing paradigm is that digital tools inform, recommend, refer, or plan and families execute. CareNavigator already determines much of what should happen. The CRP adds persistent agents that perform the administrative work required to make it happen while the family remains the decision-maker.

Agents combine LLM reasoning with the structured state in Innovation 1, explicit workflow logic, deterministic permission gates, persistent scheduling, and constrained tools. They can use APIs where available, permissions browser automation where portals require direct interaction, document preparation and extraction, email, SMS, fax, scheduling, and eventually AI-assisted voice calls. A family could authorize the system to prepare an LTSS application, assemble required financial documents, contact multiple home-care agencies with the same verified case payload, schedule transportation, follow up on post-discharge administrative tasks, or identify and activate underused nonclinical insurance benefits. Actions requiring attestation, legal authority, or consequential family choice remain human-controlled.

The same household: information returned versus work carried to outcome

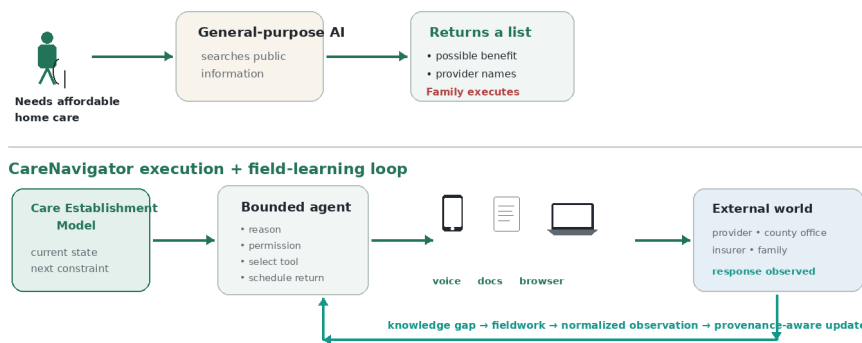


Figure 6. General-purpose AI returns information. CareNavigator executes, observes the external world, and writes verified field observations back into the model.

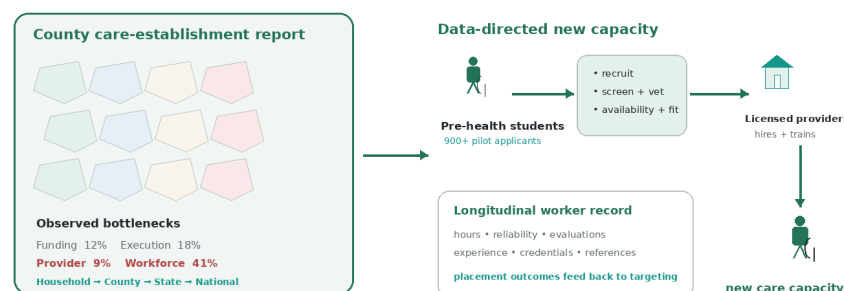
The engineering challenge is persistence across delay, incomplete information, and heterogeneous external systems. Execution follows an event-driven loop of interpret, plan, permission, act, observe, update, wait or return, and verify or escalate. Sending a form or leaving a voicemail is an event, not a successful outcome. Cases continue until the relevant endpoint is verified, declined by the family, or cannot safely proceed. Unsupported workflows and ambiguity escalate to a human navigator. CareNavigator does not diagnose, prescribe, place clinical orders, or make clinical decisions.

Authorized health information can instead be used to extract and execute administrative tasks already defined by the family, provider, or clinician.

The second novelty is that execution itself generates field knowledge. When the model encounters an unresolved operational question, the same tools can retrieve current information, inspect a portal, email or call a representative, or escalate targeted fieldwork to a human. The LLM then normalizes the returned observation into the Care Establishment Model with source, geography, time, and provenance. A provider may publicly accept a payer yet report by phone that weekend capacity is unavailable until November. That fact improves the current case and every subsequent case in the same market. More executed cases therefore expose more knowledge gaps, generate more direct observations, increase local data density, and improve future routing. This active field-learning loop is especially important in eldercare because operational truth is often local, rapidly changing, and absent from the public web.

Key Innovation 3: Data-directed creation of new caregiver capacity. Better information and execution still cannot establish care when no person is available to deliver it. Much of long-term care capacity ultimately depends on the caregiver workforce, and existing staffing channels predominantly compete for workers already circulating in the same constrained labor market. Olera instead uses the instrumented pathway to identify where workforce is actually preventing care and targets creation of new supply toward those deficits. CareNavigator can combine funded family demand, provider denials and vacancies, service type, geography, shift requirements, worker availability, placement outcomes, and external workforce indicators into a local capacity view. The first targeted supply pathway is pre-health students. They are geographically distributed through universities, replenished each academic year, often seek documented patient-facing experience, can work evenings and weekends that providers report are difficult to fill, and have career-linked incentives for reliability and continuity. Olera can reach them through advising offices, student organizations, career networks, and direct digital outreach. Our existing pilot generated more than 900 applicants, providing preliminary evidence that this is a reachable pool rather than a hypothetical source of labor.

Instrumented cases reveal where capacity fails and where new supply is most valuable



Within the CRP, Olera recruits, screens, vets, characterizes availability and fit, and refers candidates to licensed providers, which retain responsibility for hiring, training, employment, and supervision. A longitudinal worker record then accumulates verified hours, experience, credentials, provider evaluations, reliability, and references. The architecture is labor-pool agnostic even though students are the first test. The broader innovation is a repeatable mechanism for identifying an underused workforce, activating it, standardizing the information employers need, and measuring whether the resulting supply actually increases care capacity.

Figure 7. Household cases aggregate into geographic bottleneck intelligence. Caregiver Staffing is the CRP's first targeted intervention because workforce is a terminal constraint on care.

The same instrumentation has value beyond staffing. Household-level state can aggregate into county, state, and national visibility on unmet needs, funding failures, processing delays, provider deserts, workforce shortages, time to established care, and other pathway bottlenecks. That creates a new analytic layer for researchers, risk-bearing institutions, and public agencies to understand where resources are failing to reach older adults and where interventions may have the highest value. Caregiver Staffing is the first intervention because the national workforce shortage is already large and because a missing caregiver cannot be solved by better software alone.

Final End-Product: The Olera CareNavigator. The three innovations converge behind the existing CareNavigator web and mobile experience, which Phase IIB developed through repeated build-measure-learn cycles with family caregivers. The front end remains intentionally simple while the new computational and agentic infrastructure works in the background. A family leaving the hospital can be followed until needed services, transportation, and follow-up are administratively established. A family unable to afford home care can move from needs and means through aid execution, provider capacity, and service start. When workforce

is the barrier, the same system can activate Caregiver Staffing. Across use cases, the endpoint remains the one established in Significance: aid or care actually established before unmet need feeds the vicious cycle.

One CareNavigator experience. Different needs. The same endpoint.

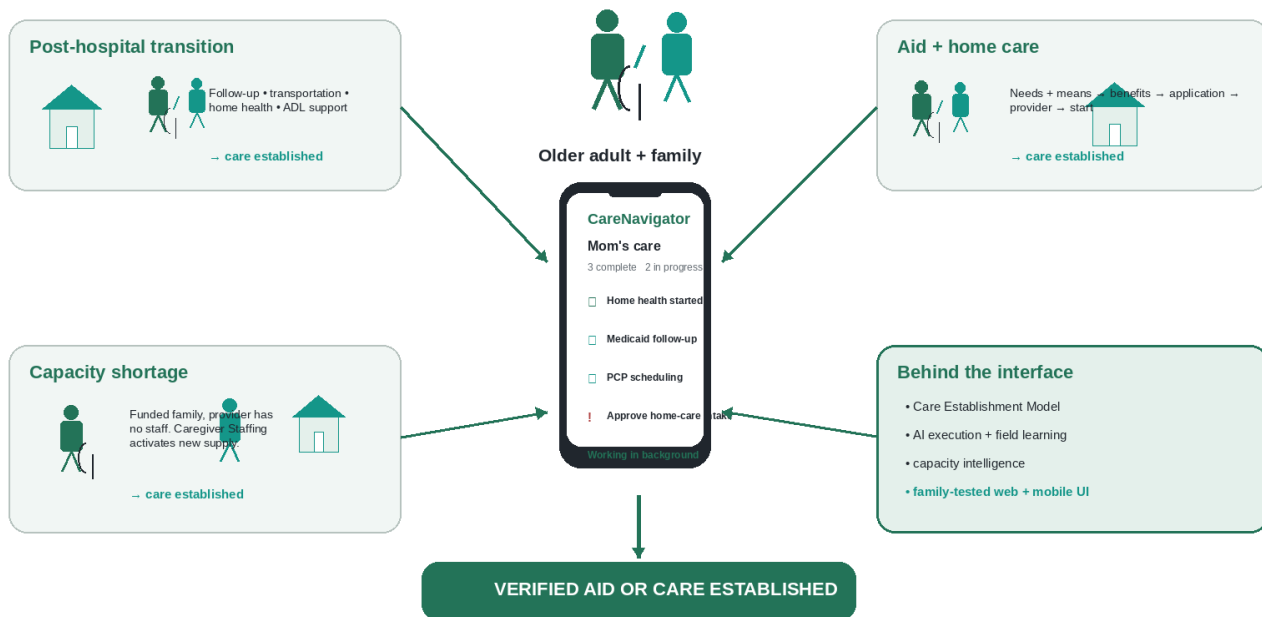


Figure 8. One CareNavigator experience supports different paths through the eldercare ecosystem while the computational model, agents, and capacity system operate behind the interface.